Homework 5: Reading a real option chain

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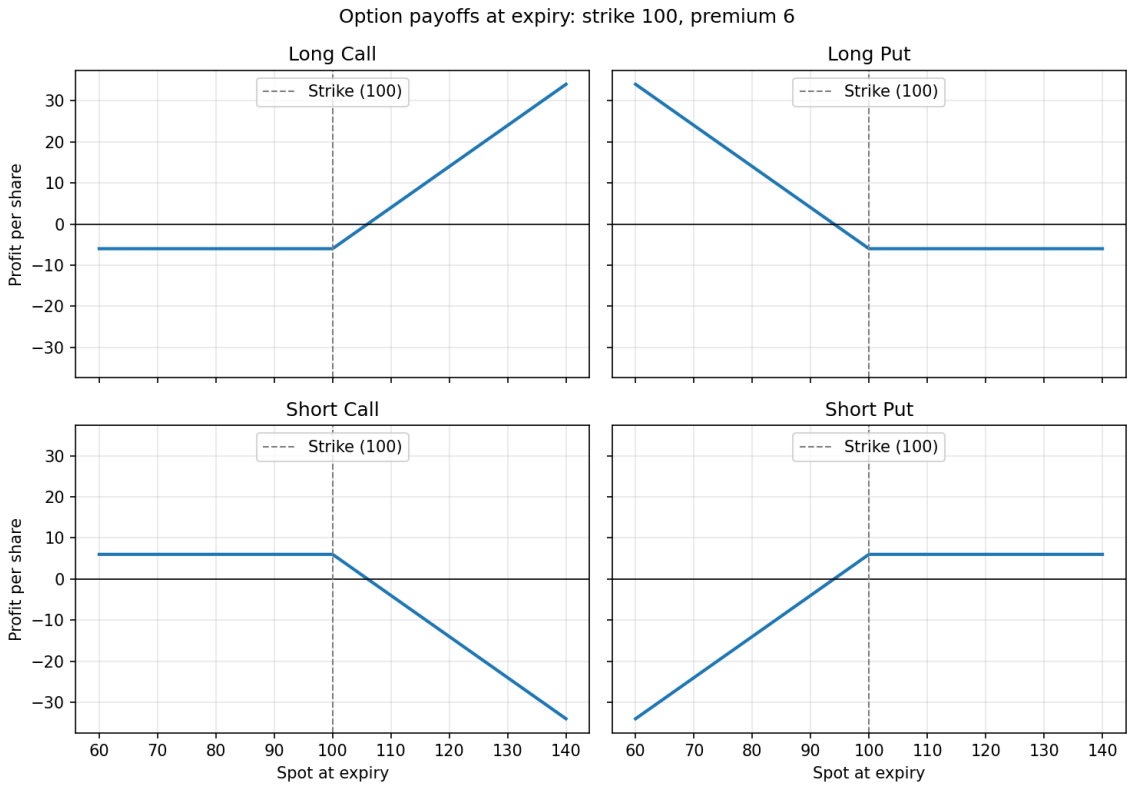
Payoffs

1. payoff(kind, strike, premium, spot)

```
strike 100, premium 6
kind            S=80   S=100   S=120
long_call      -6.00   -6.00   14.00
long_put       14.00   -6.00   -6.00
short_call      6.00    6.00  -14.00
short_put     -14.00    6.00    6.00
```

Both sign checks pass and are asserted in HW5.py: a long call at spot 80 loses exactly the 6 premium, and a short call at spot 120 loses 20 of intrinsic minus the 6 collected, so -14. Each short position is the mirror image of the matching long one, so the four rows pair up as exact negatives. The function also accepts a numpy array of spots, which Homework 6 will need for adding payoffs together. HW5.py keeps the starter's function names (intrinsic, extrinsic, payoff), and check_hw05.py passes all 14 of its checks.

2. Payoff diagrams



A live chain

All quotes are SPY options from yfinance, pulled on Sunday 25 October 2026, so they are the closing quotes from Friday 23 October. Spot is Friday's close. HW5.py picks the expiry closest

to 30 days out, and the first expiry at least six months out. Because the live chain changes every day, the exact quotes used are saved alongside the script as chain_2026-11-27.csv and chain_2027-06-17.csv.

3. Ticker, expiry, spot, chain size

```
ticker: SPY
expiry: 2026-11-27 (33 days out)
spot:   759.97
calls:  123, puts: 114
```

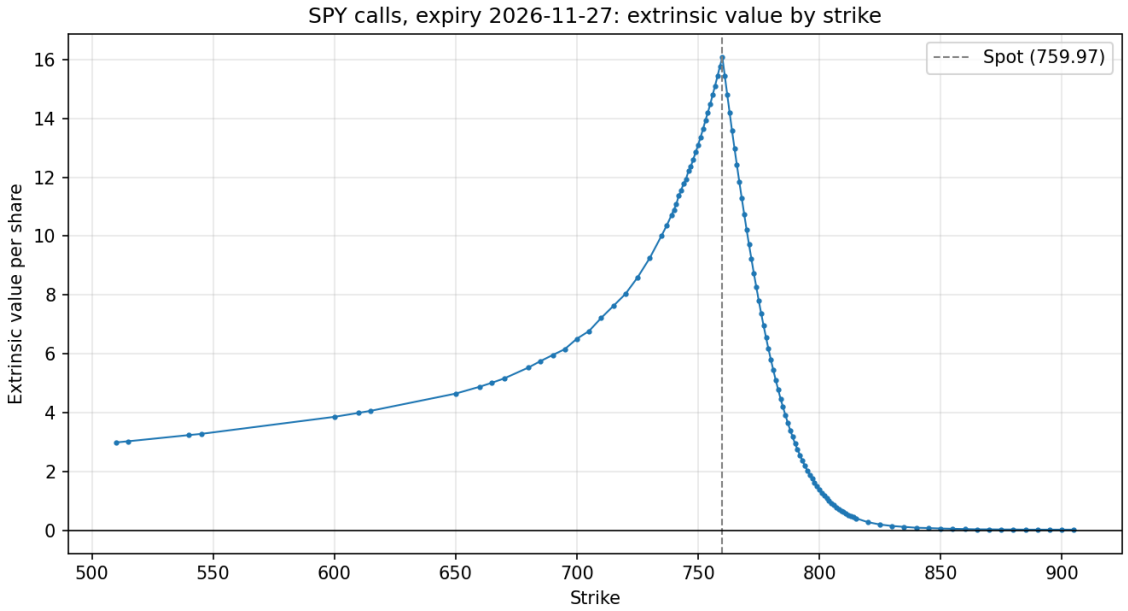
4. Mid, intrinsic and extrinsic, ten strikes nearest spot

```
2026-11-27: 119 of 123 calls kept (4 zero bid, 0 ask below intrinsic)
  strike       mid  intrinsic  extrinsic
   755.0    19.450      4.970     14.480
   756.0    18.765      3.970     14.795
   757.0    18.080      2.970     15.110
   758.0    17.405      1.970     15.435
   759.0    16.740      0.970     15.770
   760.0    16.080      0.000     16.080
   761.0    15.450      0.000     15.450
   762.0    14.815      0.000     14.815
   763.0    14.200      0.000     14.200
   764.0    13.575      0.000     13.575
```

Mid is (bid + ask) / 2, intrinsic is max(spot - strike, 0), and extrinsic is mid minus intrinsic, so intrinsic + extrinsic equals mid by construction (also asserted). The 4 calls with a zero bid were dropped.

5. Extrinsic value across the whole chain

```
largest extrinsic: 16.080 at strike 760.0
spot for comparison: 759.97
mean extrinsic for strikes <= 80% of spot: 3.275
lowest strike 510.0: extrinsic 2.985, interest on strike at 4.03% = 1.856, bid-ask spread 2.71
```



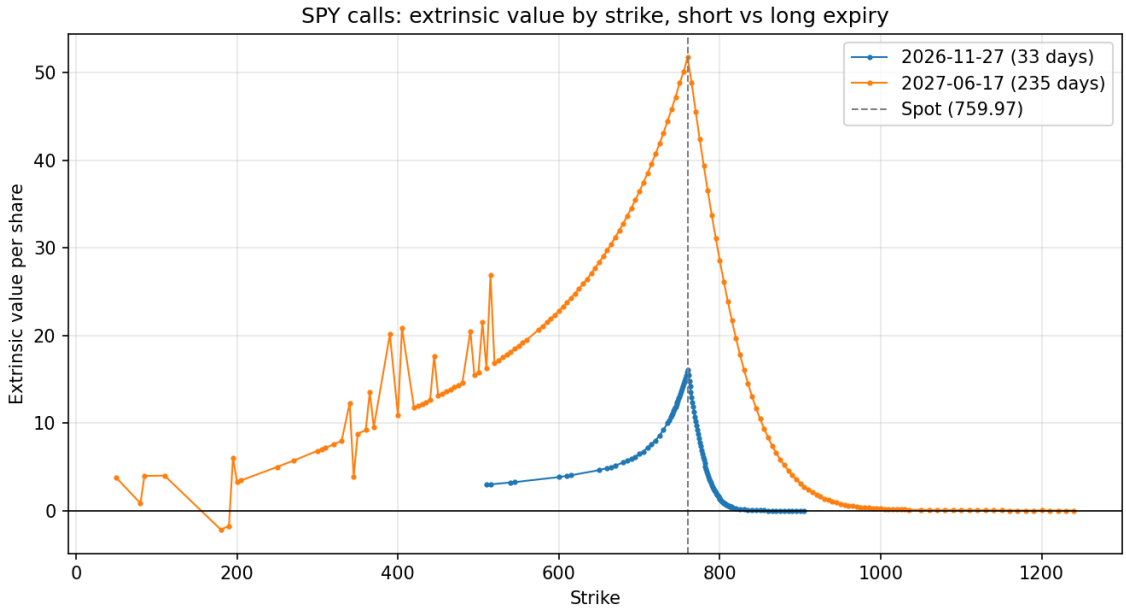
The peak is at the 760 strike, 3 cents from spot. Extrinsic value falls off on both sides, but not symmetrically. Above spot it decays to almost exactly zero by about 850. Below spot it levels off near 3 rather than reaching zero. That floor is mostly interest: someone who holds a call

pays the strike at expiry rather than today, so even a deep in-the-money call is worth about spot minus the discounted strike. At the 510 strike, 33 days of interest at the 4.03% T-bill rate is worth 1.86. The other 1.13 is less than half of that contract's 2.71 bid-ask spread, so the mid simply is not precise enough to show it. Measured against the 16.08 peak, deep in-the-money extrinsic is near zero, and what remains is the cost of money rather than the value of optionality.

6. A longer-dated expiry

```
long expiry: 2027-06-17 (235 days out)
166 of 217 calls kept (38 zero bid, 13 ask below intrinsic)
  strike       mid  intrinsic  extrinsic
   735.0    69.425     24.970     44.455
   740.0    65.780     19.970     45.810
   745.0    62.200     14.970     47.230
   750.0    58.770      9.970     48.800
   755.0    55.065      4.970     50.095
   760.0    51.755      0.000     51.755
   765.0    48.795      0.000     48.795
   770.0    45.555      0.000     45.555
   775.0    42.405      0.000     42.405
   780.0    39.405      0.000     39.405

largest extrinsic, 2026-11-27: 16.080 at strike 760.0
largest extrinsic, 2027-06-17: 51.755 at strike 760.0
ratio 3.22, sqrt of time ratio 2.67
implied vol at strike 760.0: 17.5% (2026-11-27), 21.6% (2027-06-17)
long curve above short at 57 of 57 shared strikes
```



The long-dated curve sits above the short-dated one at all 57 strikes where both expiries have a quote, and both peak at the same 760 strike. Both chains use the same underlying and the same spot from the same day.

On top of the zero-bid rule, I dropped 13 long-dated calls whose ask was below intrinsic. For example, the 120 strike was quoted at 589.00 / 593.91 against an intrinsic value of 639.97. A real market cannot sit there, because you could buy the call and exercise it at once for a 46-point profit. All 13 had zero or one contract of open interest and had last traded between

March and June, so they are stale quotes left on strikes nobody trades. Below about 520, the long-dated curve is still jagged for the same reason: these strikes are thin and their bid-ask spreads run 3 to 5 points. The 180 and 190 strikes even show slightly negative extrinsic (-2.14 and -1.72), which is well inside their 4.5-point spreads. Ignoring the spikes, the long-dated curve below spot follows the same interest-on-the-strike floor seen in question 5, which is bigger here because there are 235 days of interest instead of 33.

Written

7. Where extrinsic value is largest, and why time costs money

Extrinsic value is largest at the strike sitting right on spot, 760 here against a spot of 759.97, because that is where the outcome is most uncertain. With a 760 strike, about half the probability of where SPY ends up in November is on each side of the strike. Every path that finishes above 760 pays off, and every path that finishes below is capped at a loss of the premium, so the option holder keeps the upside and gives up the downside on a full coin flip. Move the strike away from spot and that probability mass piles up on one side. For a call struck at 510, SPY would have to fall a third in a month before the floor mattered. There is almost no probability on the wrong side of the strike, so the call just behaves like the stock, and the right to walk away is worth almost nothing. For a call struck at 900, SPY would need to rally 18% for it to pay anything. Almost none of the distribution reaches that far, so the option is worth almost nothing at all. Extrinsic value is essentially the price of the paths that could still cross the strike, and the most paths cross when the strike is closest to where the price is now.

The longer-dated curve is higher because the range of places SPY could end up keeps widening the further out you look. In 33 days, SPY usually ends up within about 5% of where it started, so only strikes near 760 have a real chance of being crossed. Over 235 days the distribution is far wider, so much more probability can still travel across any given strike, including strikes that are almost hopeless on the one-month chain. That is why the long-dated curve is above the short one everywhere and spreads out much further. The spread of outcomes grows with the square root of time, so 7.1 times as many days should give about 2.7 times the at-the-money extrinsic value. The chain shows 3.2 times as much (51.76 against 16.08). Nearly all of the gap comes from the market pricing more volatility per year over the longer window: Yahoo's implied volatility at the 760 strike is 21.6% for June against 17.5% for November, and 2.7 x 21.6 / 17.5 is about 3.3. So when you buy time, you are paying for more chances for the price to wander across your strike before the option expires, plus the use of the strike money until then. Since that spread grows only with the square root of time, each extra month buys less than the one before it.